



PDF Download
2361999.2362022.pdf
25 February 2026
Total Citations: 8
Total Downloads: 434

 Latest updates: <https://dl.acm.org/doi/10.1145/2361999.2362022>

RESEARCH-ARTICLE

A reference architecture template for software-intensive embedded systems

ULRIK EKLUND, Chalmers University of Technology, Gothenburg, Vastra Gotaland, Sweden

NIKLAS JONSSON, Volvo Car Corporation, Gothenburg, Västra Götaland, Sweden

JAN BOSCH, Chalmers University of Technology, Gothenburg, Vastra Gotaland, Sweden

ANDERS ERIKSSON, Volvo Car Corporation, Gothenburg, Västra Götaland, Sweden

Open Access Support provided by:

Volvo Car Corporation

Chalmers University of Technology

Published: 20 August 2012

Citation in BibTeX format

WICSA/ECSA '12: WICSA/ECSA 2012
Proceedings Companion Volume
August 20 - 24, 2012
Helsinki, Finland

A Reference Architecture Template for Software-Intensive Embedded Systems

Ulrik Eklund
Dept. Computer Sc. & Eng.
Chalmers Univ. of Technology
Göteborg, Sweden
ulrik.eklund@chalmers.se

Niklas Jonsson
Dept. 94121, PVD2:1
Volvo Car Corporation
Göteborg, Sweden
njonss4@volvocars.com

Anders Eriksson
Dept. 94121, PVD2:1
Volvo Car Corporation
Göteborg, Sweden
aericso5@volvocars.com

Jan Bosch
Dept. Computer Sc. & Eng.
Chalmers Univ. of Technology
Göteborg, Sweden
jan.bosch@chalmers.se

ABSTRACT

The paper presents a document template for reference architectures for the domain of software-intensive embedded systems. The template addresses the somewhat conflicting needs when documenting a reference architecture of readability, overview, details, evolution and different background of the stakeholders.

The first novel idea is the template prescribes two separate documents describing a reference architecture beside the product architecture documentation. The information in the two documents concern different stakeholders and evolve at different paces. The other novelty in describing reference architectures is to introduce a perspective of time, in the form of horizons, rather than focusing on a snapshot a release date.

In addition, a case is presented giving an example of how a reference architecture was documented according to the template, how stakeholder needs were captured, and how the different types of information was identified or recovered from existing product architectures.

Categories and Subject Descriptors

C.3 [Special-purpose and application-based systems]: Real-time and embedded systems; D.2.11 [Software engineering]: Software Architectures; D.2.7 [Software engineering]: Distribution, Maintenance, and Enhancement—Documentation

General Terms

Documentation

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, to republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee.
WICSA/ECSA 2012 August 20-24, Helsinki, Finland
Copyright 2012 ACM 978-1-4503-1568-5/12/08 ...\$15.00 ...\$15.00.

1. INTRODUCTION

An architecture description should capture the details of the implementation with the rationale and design principles behind it. This information should be accessible to a wide range of stakeholders besides the developers which creates an inherent contradiction between general readability and details concerning specific stakeholders. In the case of design principles the information also evolves on a much slower scale. One solution to this could be to capture the implementation details in a product architecture description and capture common design principles in a reference architecture.

Most reference architectures are deduced from a product software architecture to establish common objectives for a set of platforms or product families. It increases the possibility to support the same quality attributes in different products which is important to achieve consistency within the product family.

One view of what constitutes a reference architecture (RA) is that it expresses common (quality) targets for multiple platforms. This supports quality attributes getting similar penetration in multiple platforms which could be important e.g. to establish a clear brand identity.

Bass et al. defines a reference architecture as “a reference model mapped onto software elements (that cooperatively implement the functionality defined in the reference model) and the data flows between them”. [2]

Another view is that the RA is a superset, a tool box, with every possible architecture element described, which can be used in the design of a product architecture. The problem with this view is that the RA would undergo a substantial number of changes under a short period of time if the domain demands constant innovation of features. This view converges with a product line architecture [9], but perhaps with less focus on capturing variation points.

At fourth view is that a RA “capture the essence of existing architectures, and the vision of future needs and evolution to provide guidance to assist in developing new system architectures.”[5]

Inspired by Cloutier et al. [5] this paper describes the documentation of a RA by capturing the design ideas and patterns that permeates the architecture with a focus on

the design decisions that have great impact on the product family as a whole. In this sense the RA described here is a “non-structured reference architecture” according to [1], and thus falls outside of their five types of structured RA.

A RA and a product architecture differs, among many other aspects, in that the reference architecture dictates the patterns and principles to implement, i.e. what the design “*should*” be, while the product architecture instead captures the instance of the reference architecture with deviations meeting various constraints, i.e. what the design “*is*”. Another difference between a reference architecture and a product architecture is that the latter gives a coherent view of the product(s) in a project, possible to track the status of.

2. GOALS OF A REFERENCE ARCHITECTURE DESCRIPTION

There can be many goals with documenting and maintaining a reference architecture [5]. In order to limit the extent of documents and minimise the documentation maintenance the document template presented here focus on the following goals:

- Organisational consensus
- Support conformity over time
- Support consistent communication towards a wide range of stakeholders.

while deemphasizing the following goals:

- Facilitate multi-site development of different products driving differences in architecture, i.e. the RA development is not distributed.
- To be a superset of proven successful architectural solutions. The intent is not to have a toolbox of different solutions to choose from when defining a product architecture.
- The reference architecture as a formal contract.

2.1 Organisational Consensus

The consensus in the organisation about the evolution of the architecture is supported by having an architectural description that is available and easy to comprehend. The RA description concerns a wide range of stakeholders and should therefore not rely heavily on technical jargon or notation languages specific for a particular stakeholder, e.g. programmers.

2.2 Conformity over Time

The reference architecture is thought to counteract architectural erosion. This is done by capturing the fundamental ideas of the architecture and therefore facilitate adherence to these ideas. Many architecture decisions stem from realising a goal or vision of the system and it is thought that this vision must be explicated to maintain the same direction when an architecture evolves. For example it is easy to make decisions based on short-sighted business cases for individual components without understanding any cross-cutting concerns, but if the architectural ideas are marketed and widely accepted it is easier to survive such ordeals.

2.3 Consistent Communication

A RA concerns a wide variety of stakeholders with different backgrounds, some not even having a background in

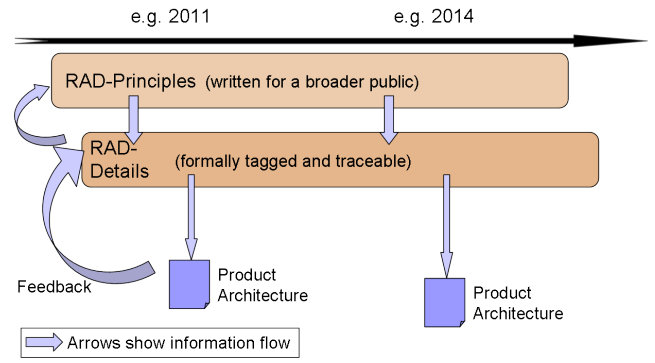


Figure 1: The two reference architecture documents and how the product architecture is realised from on them.

software development. Still the architecture, and how it is thought to evolve over time, must be conveyed to and internalised by all these stakeholders. A documented RA also supports consistency in the communication of the architecture over time.

3. DOCUMENTING A REFERENCE ARCHITECTURE

A RA description needs to address a set of somewhat conflicting needs, such as readability for all stakeholders, giving an overview of the system and provide an overview of the evolution of the architecture while also providing necessary technical details and traceability to the product architecture.

Therefore we propose describing the RA in two related documents with two different purposes and different audiences, seen in Figure 1. One capturing the essential principles and evolution of the reference architecture, written for a broad set of stakeholders. The other capturing necessary technical details, providing the foundation for the realisation of the product architecture, written mainly for architects and other developers. The first document aims to be easy to communicate and easy to digest. The documents are separated because; 1) they target different stakeholders with their concerns and need of language, 2) the information contained evolves at different paces.

The principles document is thought to be fairly stable over time, evolving slowly, possible to capture in a comprehensive manner. The details document captures the strategies and patterns that arise to meet different business needs and will be updated regularly with a faster pace compared to the principles document to reflect changes in functionality and quality attributes. The product architecture is a realisation of the abstract strategies in these two RA documents, rather than instantiated as would be the case with a product line architecture.

The template presented below for these documents are inspired by the concept of reference architectures as defined by Cloutier et al. [5]

3.1 Reference Architecture Principles

The principles document explains key elements guiding system design, goals and the roadmap towards them. This is a high level description with few details. The focus is on

the specific parts of the system that (for various reasons) need to be controlled. Note that this document does not contain a high level abstraction of the system as a whole but rather guiding principles impacting system design. A hypothetical example of technology roadmap from the automotive domain could be “The ECU architecture complies with the AUTOSAR standard and is applied to all ECUs except LIN slaves and Infotainment ECUs”.

The main audience of the document are generally everybody in a system development organisation, including developers, managers, project leaders, technology strategists, and acquirers.

This document is intended as a condensed abstract of the ideas that guides the development of the system and a guide to all involved in the development to understand the context of their daily work. The focus is to be understandable, accessible and to explain the big steps taken over time.

To make sure the information herein is usable, stakeholders should be involved, and technical solutions must be analysed, even if not presented in the document.

The design topics that need attention from a reference perspective are collected in design areas, e.g. “Connectivity” or “Propulsion” in the automotive domain. The areas are, in fact, not limited to design, but can contain processes or organisation. All design areas are subject to changes over time and those changes are mainly focused on the start of a product update cycle. These cycles are called horizons.

To explain the architectural principles the document is organised as seen in Table 1. The document starts with laying out the product vision (chapter 3) and then explains the challenges (Chapter 4) which are derived from the product vision and current technical problems. Chapter 5 describes how the design goals of the reference architecture meet the challenges. These goals are broken down in design aspects which are being elaborated on per horizon in the horizon overview (chapter 6). The design aspects are being explained more in detail in chapter 7 in the RA.

3.2 Reference Architecture Details

The Reference Architecture Details document contains the quality attributes and strategies¹ from a requirements traceability perspective and can be referenced by other design specifications.

The main audience of the document are those that develop, describe, and document product architectures, i.e. architects. The organisation of the document is seen in Table 2.

The document provides a coherent traceability between the stakeholders and their concerns in the form of quality scenarios, and the architectural strategies and patterns that promotes these qualities. The quality scenarios serve to analyse and evaluate the chosen quality attributes. The intention is that it shall be possible to evaluate the realised product architecture against the scenarios and determine if they are fulfilled or not.

Deviations affecting the architecture should be avoided, however if deviations are deemed to be necessary, they shall be verified by the architect, and documented in the product architecture document together with the rationale for the deviations.

The architectural details document is organised according to Table 2. The document starts with establishing the

¹Also called tactics by e.g. [2]

most important prerequisites for the architecture in chapter 4. The prerequisites could be defined as quality attribute scenarios and be based on e.g. the Product Vision and Challenges in the Principles document. Chapter 5 give a brief overview of a typical product and it’s design. This chapter fits better in the product architecture description, but is included here to give an overview related to legacy products. The strategies guiding the design of each product architecture are described in chapter 6 to fulfil the prerequisites in chapter 4. The strategies guide the design and development of product architectures. A hypothetical example of a strategy from the automotive domain could be “always use AUTOSAR software component description format”. There is an informal relationship between the strategies seen chapter 6 of the RA Details document and the overall design expressed in chapter 7 of the RA Principles document.

4. CASE: AUTOMOTIVE MANUFACTURER

A modern vehicle is arguably one of the most complex products today with embedded software, both in terms of conflicting requirements and subcontractor relationships [4]. A vehicle consists of 50-100 electronic control units (ECU) with deployed software, connected via different data buses. The number of possible software configurations for a single car model can exceed 10^6 .

The architects at Volvo Car Corporation felt there were more and more factors to consider when developing architectures for the embedded system, driven by the product vision of modern Volvo vehicles, e.g:

- CO² effectiveness
- Software standards, such as AUTOSAR² and GENIVI³
- New and nuanced awake states
- Different electrical hybridization levels
- Functional Safety standard[8]
- Personalization
- Possible configurations

And in a not too distant future additional factors in the product vision include:

- Collaborative driving features e.g. Platooning⁴
- Car-to-car and car-to-infrastructure communication with authentication
- Customer specific software “apps”
- Remote software upgrades
- COTS software
- Vehicle integration with customer nomadic devices
- Electronic Horizon, Driver Assessment
- Post-crash functionality

Together with most of the innovations in a vehicle is realised with software, these trends lead to the following challenges:

- More complex solutions
- More functions utilizing the same infrastructure, both on-board and off-board
- More teams to coordinate and integrate
- Specialized architects for chassis, body & comfort, infotainment, etc.

²<http://www.autosar.org>

³<http://www.genivi.org>

⁴<http://www.sartre-project.eu>

Table 1: The chapters of the reference architecture principles document.

Chapter	Heading	Content
1.	Introduction	Definition of the scope of the RA, the principles document, the intended audience and it's relationship to other documents .
2.	Reading guideline	The reading guideline contains information similar to what is presented in Section 3.1.
3.	Product Vision	The product vision is based on the company vision to the extent communicated. It is mainly the architects' vision, based on reasonable assumptions founded in present knowledge, of what is needed to achieve during the life-time of the product platform.
4.	Challenges	This chapter defines architectural challenges that must be met to support the product vision. Here it is also possible to incorporate other challenges not originating from the visions, which tend to focus on performance and functionality. The challenges can e.g. originate from known problems with present implementation or constraints originating from technology.
5.	The Architectural Goals	This chapter defines the design goals of the architecture that meets the challenges. They need to be reached in the long term in order to meet the product vision.
6.	The Horizons	The number of changes in the architecture is not seen as being manageable to reach in a single step. The evolution is therefore split into a number of horizons. The horizons are matched to the product plan and to the organizational capability. The details are elaborated in chapter 7, while this chapter allows for a broader reasoning across design areas and elaborate on the evolutionary steps of the product(s).
7.	Reference Architecture Design	In this chapter the key design aspects are elaborated for each horizon, with more details closer in time and more speculative further away in time. The design aspects are selected as those having a large influence on the final system and/or will change dramatically. They do not have a clear separation of concern, but are instead chosen to give an accessible understanding of the key aspects of the system.

Table 2: The chapters of the reference architecture details document.

Chapter	Heading	Content
1.	Document change log	
2.	Introduction	Information about what the purpose is of this document, who should read it, and what relationship it has to other documents, e.g. the RA principles and the product architecture documents as seen in Figure 1.
3.	Terminology	Definitions and acronyms used.
4.	Input requirements	This chapter describes the identified, considered stakeholders to the system, and their concerns. With concerns we mean aspects that are critical or otherwise important to one or more stakeholders, for example given as requirements to fulfil, e.g. both functional requirements and non-functional requirements/quality attribute scenarios. The prioritisation of the quality attributes is also captured here.
5.	System description	A system overview supporting understanding of the products to be built.
6.	Architecture strategies	This chapter defines the architecture strategies that should be followed during development in order to establish a common architecture among various products. The strategies are generic and shall be applied when the product architecture is realised.

All this cause architecture erosion, and the architects decided to counteract this by describing the embedded reference architecture for future vehicle platforms. The scope of the reference architecture is the complete electrical systems, including software, and encompasses e.g. the basic software on each ECU, choice of computer network technologies and topologies, subsystem interaction and electrical system modes.

This effort was also aligned with the complete vehicle strategy for the forthcoming platform (which include chassis, engine and body beyond electronics and software):

“Volvo Car Corporation’s technological future is based on a Scalable Product Architecture (SPA) which in principle means that most Volvo models can be built on the same production line irrespective of vehicle size and complexity. When the first model built on SPA is launched in 2014 - the new XC90 - about 90 per cent of its components will be new and unique.”⁵

The main difference to previous attempts at Volvo Cars to define a reference architecture [6, 10] is the evolution of the system over time as a key part of the description, instead of the RA just being a snapshot.

4.1 Stakeholder study

In order to identify the information need among various stakeholders at Volvo Cars a study was performed with architects and purposefully selected persons having a good knowledge of and interfacing with the present product architecture. In total 21 persons participated in the study, participating in interviews. Out of these 10 persons, architects or others concerned with cross-cutting issues, also answered a questionnaire aiming to show the spread in answers to the interview questions.

The interviews started with general questions aimed at understanding the need of information at Volvo Cars. The interview ended with more specific questions on how the information should look like, possible receivers of information, etc.

One thing noted in the study was that depending on the interviewees’ view of a RA they imagined different readers and different types of information documented, and thus influenced the final proposal of separating architectural information between two RA documents, as described in Section 3. This is reflected in Figure 2 since the questionnaire was preceded by a general discussion about RA.

Further results from the study, with the questionnaire results seen in Figure 3, were that a RA description should contain strategies, quality attributes and the rationale that underlies the architecture. The level of detail varied among the respondents, as seen in in Figure 3 since the architects varied in their opinion of how the architecture should look like. The study claimed a RA should emphasize validation over verification, i.e. it is more important to have the right architecture than to describe in such a way the resulting system can be tested to adhere to the architecture.

Another result was a need of a RA that can easily be communicated with explained patterns and principles on a high level, but not on the expense of architectural structures and the ability to analyse them. At the same time there is a

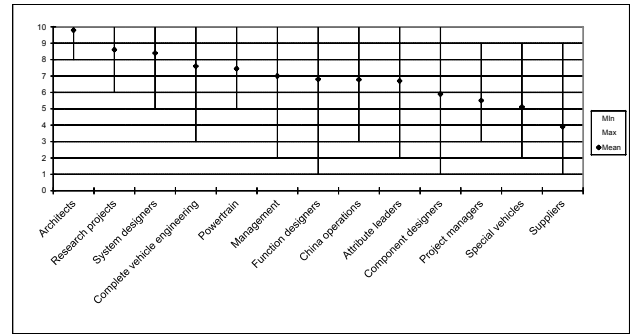


Figure 2: Which stakeholders at Volvo Cars a reference architecture should cater to, according to the results from the questionnaire. The higher the number, the more interested the stakeholder was thought to be of a RA.

need to preserve consistency over the platform and support traceability, it must be possible to trace from requirements to solution that the architecture supports required attributes.

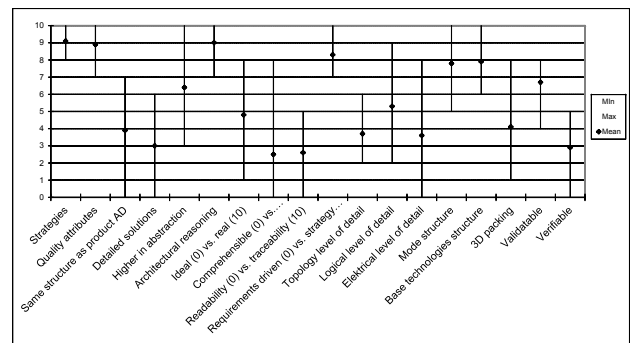


Figure 3: The form and content of a reference architecture description according to the results from the questionnaire. The higher the number, the more important the type of information was thought to include in a RA.

The result from the study was summarised by Volvo Cars: The RA description should facilitate continuity over time. It should support understanding among stakeholders and discussion of patterns. It should be easy to work with as a tool for communication.

4.2 Vehicle Reference Architecture Principles

The scope of the reference architecture is the electrical system inside the vehicle, encompassing all subsystems controlled by software and electronics, such as the engine management system, the locking systems, the instrument cluster and the infotainment system.

Since the RA description at Volvo Cars contains the future technology cycle plan it is highly sensitive from a business perspective, therefore it is not possible to give a comprehensive overview of the architecture. Instead we will select

⁵<https://www.media.volvocars.com/>

a few small parts based on already public information⁶ and give a general outline of how those parts were documented according to the template in Section 3.

The RA principles document at Volvo Cars is not very verbose, presently only 31 pages, to support readability and accessibility to a wide range of stakeholders.

4.2.1 Introduction

This reference architecture document is intended for all stakeholders at Volvo Cars and all employees working with the electrical system.

4.2.2 Reading Guideline

The reading guideline of the RA at Volvo Cars contains information similar to what is presented in Section 3.1.

4.2.3 Vision

Based on the new global corporate and brand strategy - “Designed Around You” a vision for the electrical system was defined by the architects covering 6 areas. One of these describes the vision for the service and aftermarket. In the present strategy called “Service 2.0” it is stated

Gratuitous updates: At Volvo we always improve the software in our cars. Therefore updates to the latest version is always included in Volvo Service 2.0⁷

The example is based on an already implemented vision due to the sensitivity of making the future product vision public, but shows the general reasoning throughout the Volvo RA documents.

4.2.4 Challenges

Nine challenge areas were identified that needs to be addressed in order to meet the vision, e.g. the growth of functional complexity.

Challenges when implementing the software upgrade vision are:

- Minimise the necessary electrical energy consumption, a car on standby to receive software upgrades consumes electric power.
- Having a cost effective solution, the on-board hardware and the infrastructure must support deployment to hundreds of thousands of cars.
- Not reducing the availability of the car to the user, the upgrade must be done in a timely manner without affecting the normal operation of the vehicle by the customer.

4.2.5 Architecture Design Goals

This is where the goals for the design of the architecture are defined. These need to be reached by the year 2020 in order to solve the challenges and meeting the product vision. The goals are presently defined for 7 areas, with Logic, Networks & topology, ECU architecture, and Diagnostic services being four of them.

The particular design goal about the actual solution for software upgrades can unfortunately not be disclosed. There

are two fundamental possibilities, either the in-vehicle solution is distributed with each ECU keeping track of which software goes into it when the car is connected, or the solution is centralized with a central master ECU distributing software to the right target ECU.

4.2.6 Horizons

The RA defines different time horizons for the next platform and what it will be technically capable of in terms of software updates, e.g:

2012-2014: Fully automatic download of latest software versions when servicing the vehicle at the workshop.

2015-2018: Remote download of new software in running vehicle.

4.2.7 Reference Architecture Design

The principal design of the reference architecture is defined for 9 “topics” or informal views⁸, with three examples of them being Logical structure, Electrical supply and distribution, and PMT (Process, methods & tools). This chapter describes how to architecturally address the problems within the topic/view.

For the software upgrades it describes both how to logically solve the upgrade problem and what components need to be active when software is downloaded to the vehicle.

4.3 Vehicle Reference Architecture Details

The Details document at Volvo Cars is presently 54 pages, with about 30 of these detailing the architectural strategies for the in-vehicle electrical system.

4.3.1 Input Requirements

The document describes 19 different stakeholders in 6 categories, with end users of the system (i.e. drivers and passengers of the vehicle), and developers of the electrical system being two of the categories. Some of the stakeholders can be seen in Figure 2.

The document defines 20 quality attributes, seen in Figure 4, and list which of these are most prioritised. This does not mean that non-prioritised quality attributes will not be fulfilled to satisfactory level, just that when balancing architectural decisions, the prioritised quality attributes will carry more weight.

For each of the quality attributes one or more scenarios are defined which the finished system should fulfil. The rationale for the quality attributes connects to what is defined in the RA principles document, e.g. in the Vision chapter. A simplified example of a quality scenario regarding reprogramming at workshops:

The average re-programming time for a vehicle in a workshop should not be longer than 6 minutes and in worst case no longer than 48 minutes.

Rationale: The after sales need prohibits long programming times, regardless of chosen programming technology.

Example: Some ECUs already solve this by programming done via USB and not via CAN in the CARB connector which is the standard method.

⁶<http://www.volvocars.com>

⁷http://www.volvocars.com/se/sales-services/services/service2_0/

⁸9 challenges and 9 “topics” is a coincidence, they do not have a 1-to-1 relationship.

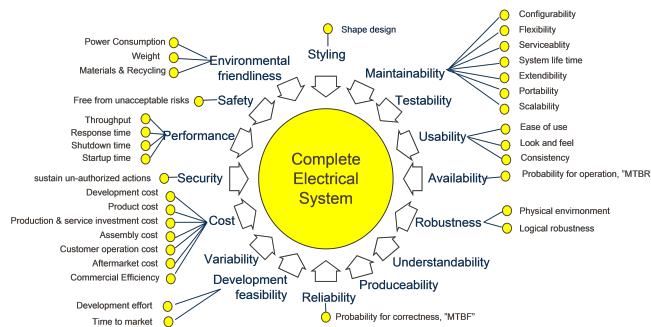


Figure 4: Quality Attributes relevant for embedded automotive systems.

4.3.2 System description

The document gives a brief overview of the system, e.g. what architectural style the architecture follows, and what views one can expect in the product architecture description, which also contains the formal definitions of the viewpoints.

4.3.3 Strategy

This chapter presently defines 32 strategies that should be followed during development in order to establish a common architecture. Most of them are either found in present product architecture documents or rediscovered when analysing present designs. Capturing common strategies also means that each product architecture description becomes more succinct.

The strategies are generic and shall be applied when realising every product architecture, if deviations are occurring, they shall be verified by the architect, and documented in the project specific architectural document together with the rationale for the deviations.

For the software upgrade a small part of a strategy says

All ECUs shall be reprogrammable: The ECU will be possible to reprogram without removal or the use of ECU specific tool.

The rationale for this strategy says it aims to improve Serviceability and Updatability (as part of the system life time) from Figure 4, and supports the defined scenarios for these quality attributes.

4.4 Case Discussion

It was difficult for the engineers at Volvo Cars to find concrete cases on reference architectures in literature. Found examples either focused on what a reference architecture is or how to design a RA [3, 5, 7, 1]. Almost nothing found contained actual examples, which could be due to the sensitivity of the information, same as for Volvo Cars. Most of the template described in Section 3 is therefore based on experiences from Volvo Cars over a number of years together with some literature [5].

In previous attempts at Volvo Cars to capture the platform architecture [6] the documentation followed one-document-fits-everybody, in the sense of trying to describe all content appropriate to multiple stakeholders. In practice this meant that for each meeting with each stakeholder a significant amount of time was spent to compile information tailored to each stakeholder, based on the architects' internalised knowledge.

The decision to increase the number of architecture documents; RA principles, RA details, and the product architecture, would potentially mean that the documentation maintenance would increase, but the fact that these documents evolve at quite different paces made it worthwhile.

The other novelty was to introduce a perspective of time, in the form of horizons, rather than focusing on a snapshot for a single vehicle or family of vehicles at a production date.

5. CONCLUSION

The paper presented a template for reference architectures in two documents together with a case of a software-intensive embedded system.

The details of the template is described, with the first document explaining the key elements guiding system design with the goals and the roadmap towards them expressed as different time horizons. The second document contains the quality attributes, architectural strategies and patterns that guide the development of product architectures. The division in two documents addresses the somewhat conflicting needs of readability, overview, details, evolution and different background of the stakeholders.

The presented case from an automotive manufacturer gave an example of how a reference architecture was documented according to each heading of the template, how stakeholder needs were captured, and how the different types of information was identified or recovered from existing product architectures.

Acknowledgements

This work has been financially supported by Volvo Car Corporation and the Swedish Agency for Innovation Systems (VINNOVA).

6. REFERENCES

- [1] S. Angelov, P. Grefen, and D. Greefhorst. A framework for analysis and design of software reference architectures. *Information and Software Technology*, 54(4):417–431, Apr. 2012.
- [2] L. Bass, P. Clements, and R. Kazman. *Software Architecture in Practice*. SEI Series in Software Engineering. Addison-Wesley, 2nd edition, 2003.
- [3] J. Bayer, T. Forster, D. Ganesan, J. Girard, I. John, J. Knodel, R. Kolb, and D. Muthig. Definition of reference architectures based on existing systems. Technical Report 034.04/E, Fraunhofer Institut Experimentelles Software Engineering, 2004.
- [4] M. Broy. Challenges in automotive software engineering. In *Proceedings of the International Conference on Software Engineering*, pages 33–42, Shanghai, China, 2006. ACM.
- [5] R. Cloutier, G. Muller, D. Verma, R. Nilchiani, E. Hole, and M. Bone. The concept of reference architectures. *Systems Engineering*, 13(1):14–27, Jan. 2009.
- [6] U. Eklund, O. Askerdal, J. Granholm, A. Alminger, and J. Axelsson. Experience of introducing reference architectures in the development of automotive electronic systems. In *Proceedings of the international workshop on Software engineering for automotive systems*, volume 30, pages 1–6, St Louis, Mo, USA, 2005. ACM.

- [7] M. Galster and P. Avgeriou. Empirically-grounded reference architectures: a proposal. In *Proceedings of the joint ACM SIGSOFT conferences on the Quality of Software Architectures and on Architecting Critical Systems*, page 153–158, Boulder, CO, USA, 2011. ACM.
- [8] ISO. 26262: Road vehicles - functional safety, 2011.
- [9] E. Y. Nakagawa, P. O. Antonino, and M. Becker. Reference architecture and product line architecture: A subtle but critical difference. In *Proceedings of the European Conference on Software Architectures*, volume 6903 of *Lecture Notes in Computer Science*, pages 207–211. Springer, 2011.
- [10] D. Selin, H. Svensson, P. Sundsten, A. Wikström, and U. Eklund. A reference architecture for infotainment systems. In *Proceedings of Convergence 2006*, volume 2006-21-0013, Detroit, MI, USA,, Oct. 2006. SAE.